

1. (4)

A	B	C	$A(B \oplus C)$	$AB \oplus AC$
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

(5)

A	B	$A \oplus \bar{B}$	$A \odot B$	$A \oplus B \oplus 1$
0	0	1	1	1
0	1	0	0	0
1	0	0	0	0
1	1	1	1	1

2. (1) $F = [(AB + C)D + \bar{E}]BC$

$$\bar{F} = [(\bar{A} + \bar{B})\bar{C} + \bar{D}]E + \bar{B} + \bar{C}$$

$$F_d = [(A+B)C + D]\bar{E} + B + C$$

(2) $F = AB + (\bar{A} + C)(C + \bar{D}E)$

$$\bar{F} = (\bar{A} + \bar{B})[\bar{A}\bar{C} + \bar{C}(D + \bar{E})]$$

$$F_d = (A+B)[\bar{A}C + C(\bar{D} + \bar{E})]$$

3. (1) $A \oplus B \oplus C = (A \oplus B) \oplus C = A \odot \bar{B} \oplus C = A \odot (\bar{B} \oplus C) = A \odot B \odot C$

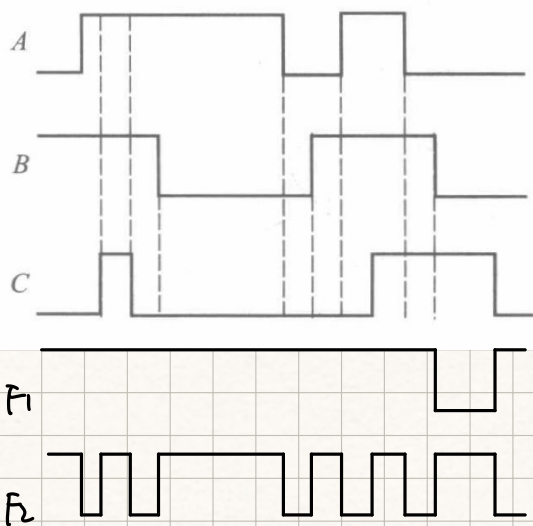
(2) $\bar{A} \oplus B \oplus C = A \odot B \oplus C = \overline{A \oplus B \oplus C}$

$$\overline{A \oplus B \oplus C} = \bar{A} \odot \bar{B} \oplus C = \bar{A} \odot \bar{B} \odot \bar{C} = \overline{A \odot B \odot C}$$

4. (1) $F_1 = \overline{(A+B)(B+\bar{C})}$ $F_2 = A \oplus B \oplus C$

A	B	C	F_1	F_2
0	0	0	1	0
0	0	1	0	1
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

(2)



3. (1) $F_1 = \sum m(4, 5, 6, 7) = \prod M(0, 1, 2, 3).$

$F_2 = \sum m(3, 5, 6, 7) = \prod M(0, 1, 2, 4).$

(2)

AB \ C	00	01	11	10
0			1	1
1			1	1

$F_1 = A = A.$

AB \ C	00	01	11	10
0			1	
1		1	1	1

$F_2 = AB + BC + AC$

AB \ C	00	01	11	10
0	0	0		0
1	0			

$F_2 = (A+B)(A+C)(B+C).$

7. (1) $F = \bar{A}\bar{C} + \bar{A}\bar{B} + BC + \bar{A}\bar{C}\bar{D}$

$= \bar{A}\bar{C} + \bar{A}\bar{B} + BC$

$= \bar{A}\bar{B}C + BC$

$= \bar{A} + BC$

(5) $F = A\bar{B}(\bar{A}CD + \overline{AD + \bar{B}\bar{C}})(\bar{A} + B)$

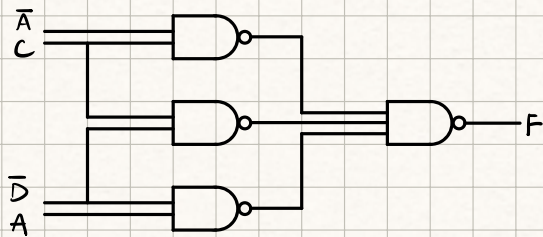
$= A\bar{B}(\bar{A}CD + \overline{AD + \bar{B}\bar{C}})\bar{A}\bar{B}$

$= 0$

8. (2) $F = \sum m(2, 3, 6, 7, 8, 10, 12, 14)$.

AB \ CD	00	01	11	10
00			1	1
01				
11	1	1		
10	1	1	1	1

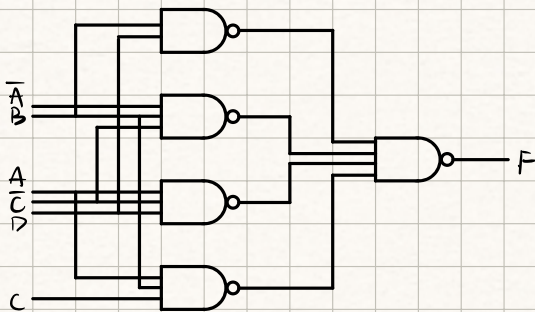
$$F = \bar{A}C + A\bar{D} + C\bar{D} = \overline{(\bar{A}\bar{C})(\bar{A}\bar{D})(\bar{C}\bar{D})}$$



(4) $F = \prod M(0, 1, 2, 3, 6, 8, 10, 11, 12) = \sum m(4, 5, 7, 9, 13, 14, 15)$

AB \ CD	00	01	11	10
00		1		
01		1	1	1
11		1	1	
10			1	

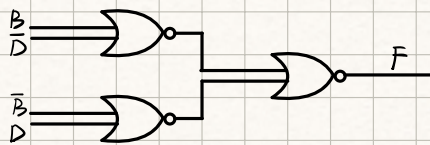
$$F = BD + \bar{A}\bar{B}\bar{C} + A\bar{C}\bar{D} + AB\bar{C} = \overline{(\bar{B}\bar{D})(\bar{A}\bar{B}\bar{C})(A\bar{C}\bar{D})(AB\bar{C})}$$



9. (1) $F = \sum m(0, 2, 5, 7, 8, 10, 13, 15) = \prod M(1, 3, 4, 6, 9, 11, 12, 14)$.

AB \ CD	00	01	11	10
00		0	0	
01	0			0
11	0			0
10		0	0	

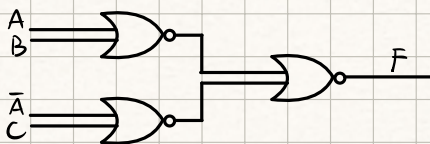
$$F = (\bar{B} + D)(B + \bar{D}) = \overline{(\bar{B} + D)(B + \bar{D})}$$



(5) $F = (A + B)(A + B + C)(\bar{A} + C)(B + C + D)$.

AB \ CD	00	01	11	10
00	0		0	0
01	0		0	0
11	0			
10	0			

$$F = (A + B)(\bar{A} + C) = \overline{(\bar{A} + B)(\bar{A} + C)}$$



11. (2)

A	B	C	$A \oplus B$	$B \oplus C$	F_C
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	1
1	1	0	0	1	0
1	1	1	0	0	1

$$F_C = \sum m(0, 2, 5, 7) = \prod M(1, 3, 4, 6)$$

12. (1) $F = \sum m(1, 3, 6, 8, 11, 14) + \sum d(2, 4, 5, 13, 15)$

AB \ CD	00	01	11	10
00		X		1
01	1	X	X	
11	1		X	1
10	X	1	1	

$$F = \bar{A}\bar{B}D + B\bar{C}\bar{D} + ACD + A\bar{B}\bar{C}\bar{D} = (\bar{A} + \bar{B} + D)(B + C + \bar{D})(A + C + D)(A + \bar{B} + \bar{C} + \bar{D})$$

(2) $F = \prod M(0, 1, 4, 7, 9, 10, 13) \cdot \prod d(2, 5, 8, 12, 15)$

AB \ CD	00	01	11	10
00	0	0	X	X
01	0	X	0	0
11		0	X	
10	X			0

$$F = C(\bar{B} + \bar{D})(B + D) = C + \bar{B}\bar{D} + BD$$

(5) $\begin{cases} F = (A \oplus B)\bar{C}\bar{D} + \bar{A}B\bar{C} + \bar{A}\bar{C}D = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C} + \bar{A}\bar{C}D \\ AB + CD = 0 \end{cases}$

AB \ CD	00	01	11	10
00			X	
01		0	X	0
11	X	X	X	0
10			0	0

$$F = (\bar{B} + \bar{D})(\bar{A} + \bar{D})(\bar{A} + \bar{C}) = \bar{B}\bar{D} + \bar{A}\bar{D} + \bar{A}\bar{C}$$

6. (1) $F_1 = \overline{A\overline{B}} + \overline{\overline{A}B} + C$ $F_3 = \overline{A+B+\overline{A}\cdot\overline{C}}$

(2) $F_1 = \sum m(1,3,5,7)$ $F_3 = \sum m(0).$